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① Let G be a group of order pq, where p and q are distinct primes. prove that G is abelian.

Answer: False.

Reason: Counter example S_3, $|S_3| = 6 = 2 \cdot 3$, but S_3 is non-abelian.

② prove that if G is a group of order p^2, where p is prime, then G is abelian if and only if it has subgroups of order p.

Answer: False.

Reason: Every group of order p^2 is abelian. There are exactly two types: C_{p^2} and $C_p \times C_p$? There correct equivalance is $G \cong C_p \times C_p \Leftrightarrow G$ has $p+1$ subgroups of order p", Both groups are abelian.

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③ Let G be a finite group and it be a proper subgroup of G, prove that the union of all conjugates of H cannot be equal be G.

Answer: False.

Reason: For Finite G the union of conjugates of a proper subgroup cannot be cover G. A finite group cannot be a union of finitely many proper subgroups.

⑤ Prove that in any group G, the set of elements of finite order forms a subgroup of G.

Answer: False.

Reason: In abelian group, yes but not always,

Example: Infinite dihedral group two reflections (order 2) multiply to a rotation of infinite order, so closure fails.

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(6) Let G be a finite group and p be the smallest prime dividing $|G|$. Prove that any subgroup of index p in G is normal.

Answer: True

Reason: Action on cosets gives homomorphism into S_p; using smallest prime property the image must force the subgroup to be normal.

(9) Let G be a finite group and p be a prime number. If G has exactly one subgroup of order p^k for each $k \leq n$. Where p^r divides $|G|$, prove that G has a normal sylow p subgroup.

Answer: True

Reason: unique subgroup of order p^r is the unique sylow p-subgroup and uniqueness implies normality.